

Phase 1: Data Ingestion, Privacy Scrubbing, and LODO Partitioning

Corpus Ingestion and Filtering

- Window: $t_0 - 24\text{h}$ to $+72\text{h}$ (PAGASA)
- 6 Philippine typhoons (2020–2024)
- Tweet hydration and deduping
- FastText LID and CMI screening

Privacy and Gold Annotation

- DPA 2012 (RA 10173) PII scrubbing
- Cryptographic SHA-256 seal
- Five-step consensus adjudication
- Acceptance: $F1 \geq .75, \kappa \geq .60$

LODO Partitioning and Audit

- 6 LODO disaster folds ($K = 6$)
- 15% stratified dev split (fixed)
- Out-of-domain holdout isolation
- Class support audit per stratum

Frozen Datasets and Leave-One-Disaster-Out Folds

Phase 2: Model Training, Multi-Task Balancing, and Telemetry Extraction

Pre-Flight and Baselines

- Workstation environment checks
- Pinned seeds: $s \in \{42, \dots, 1011\}$
- Single-task baselines (3 heads)
- Baseline variance tolerance logging

Multi-Task Balancing

- Shared ModernBERT encoder stack
- Uniform loss weighting baseline
- Tuned static loss weighting
- Gradient surgery via PCGrad

Gradient Dynamics and Safety

- Gradient cosine: $\cos(\mathbf{g}_i, \mathbf{g}_j)$
- Gradient conflict rate tracking
- Loss divergence auto-recovery
- Checkpoint frozen at peak dev F1

Frozen Checkpoints and Gradient Telemetry

Phase 3: Inference Profiling and Hardware Testbed Benchmarking

Testbed Workload Setup

- Dedicated RTX 4060 Laptop GPU
- Batch size $b = 1$ emergency triage
- Unmeasured warm-up iterations
- CUDA kernel steady-state execution

High-Resolution Measurement

- Monotonic latency timers
- CUDA synchronization barriers
- Framework peak GPU VRAM tracking
- Process host RAM RSS sampling

Serving Architecture Trials

- Unified single-model serving
- Sequential head execution trials
- Concurrent head execution trials
- Latency vs. memory trade-offs

Empirical Evaluation Metrics and Hardware Telemetry Logs

Phase 4: Statistical Hypothesis Testing, Telemetry Schemas, and Dissemination

Automated Telemetry Logs

- JSON run manifests and hashes
- CSV training loss and dev F1 logs
- CSV checkpoint metric logs
- CSV hardware profiling logs

Statistical Hypothesis Testing

- Seed averaging: $N = K = 6$
- Paired permutation ($2^6 = 64$)
- Permutation $p_{\min} = .015625$
- Stratified bootstrap ($B = 10,000$)

Open Research Dissemination

- Anonymized tweet ID ground-truth set
- Dataset documentation sheets
- Standardized model and system cards
- Public open-source reproduction code